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|--------------|----------------------|--------------------------|
| Civics Group | A Level Index Number | Name (use BLOCK LETTERS) |
|--------------|----------------------|--------------------------|

**H2**

**ST. ANDREW'S JUNIOR COLLEGE  
2021 JC2 PRELIMINARY EXAMINATIONS**

**H2 BIOLOGY****9744/03****Paper 3**

Thursday

16<sup>th</sup> September 2021Total 2 hours for  
both Set A & B**READ THESE INSTRUCTIONS FIRST**

Write your name, civics group and index number on all the work you hand in.

Write in dark blue or black pen on both sides of the paper.

You may use a soft pencil for any diagram, graph or rough working.

Do not use staples, paper clips, highlighters, glue or correction fluid.

**(Structured Question 1)**

Answer **all** questions.

Write your answers in the spaces provided on the question paper.

All working for numerical answers must be shown.

| Conceptual error<br>(C) | Data Quoting<br>(D) | Expression<br>(E) | Misreading the<br>question (Q) |
|-------------------------|---------------------|-------------------|--------------------------------|
|                         |                     |                   |                                |

| For Examiners' Use |            |
|--------------------|------------|
| Set A              |            |
| 1                  | /35        |
| Set B              |            |
| 2                  | /5         |
| 3                  | /10        |
| 4 or 5             | /25        |
| <b>Total</b>       | <b>/75</b> |

This document consists of **12** printed pages.

**[Turn over]**

*Answer **all** questions.*

### **QUESTION 1**

Many diseases are linked to components of the immune system.

T-cell lymphoma is a rare form of cancerous lymphoma affecting T lymphocytes. Lymphoma arises mainly from the uncontrolled proliferation of T lymphocytes and can become cancerous.

**(a)** Cancer is described as a multi-step process. Explain how mutations in two main groups of critical genes may lead to uncontrolled proliferation in T lymphocytes.

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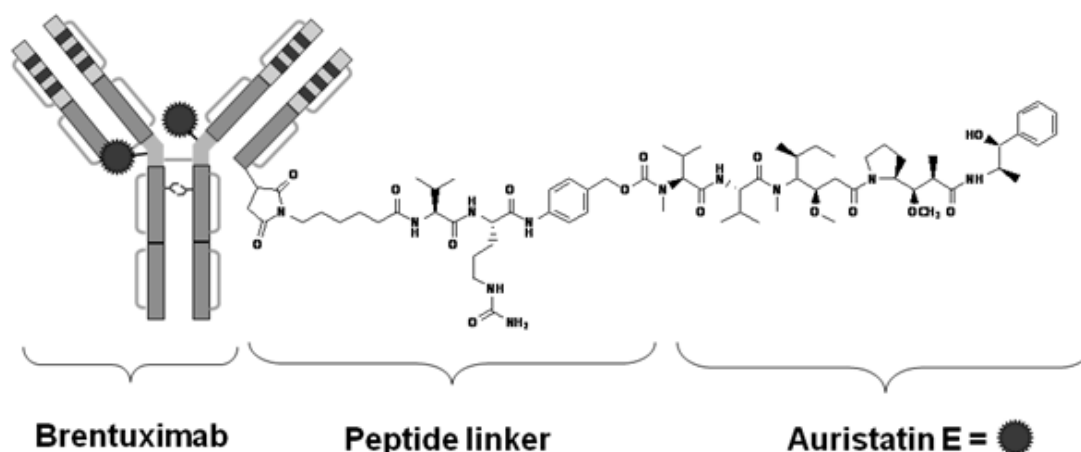
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.....[4]

- (b) Possible treatment for lymphoma might include antibodies attached to chemotherapy drugs (an antibody-drug conjugate). For example, Brentuximab vedotin is an antibody attached to a chemotherapy drug.

This antibody specifically recognises CD30 found on lymphoma cells and acts like a “homing” signal, bringing the chemotherapy drug to lymphoma cells, where it enters the cells and kills them.

Fig. 1.1 shows the structure of this antibody-drug conjugate and how the drug is attached to the antibody.



Modified from [http://www.pharmatimes.com/Article/13-05-15/FDA\\_to\\_review\\_extended\\_use\\_of\\_Seattle\\_Genetics\\_Adcetris.aspx](http://www.pharmatimes.com/Article/13-05-15/FDA_to_review_extended_use_of_Seattle_Genetics_Adcetris.aspx)

Fig .1.1

Auristatin E is an anti-mitotic agent which inhibits cell division by blocking the polymerisation of tubulin, a protein monomer of spindle fibres.

- (i) With reference to Fig 1.1, identify the region on the antibody that helps the antibody to achieve its “homing” function. Explain how this “homing” is achieved.

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 .....  
 .....[2]

- (ii) Suggest how Auristatin E can help to curb T cell lymphoma.

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 .....[2]

An investigation was conducted to determine the effectiveness of Brentuximab vedotin (antibody and auristatin E conjugate) treatment in humans.

A lymphoma cell line was incubated *in vitro* with Brentuximab vedotin (antibody and auristatin E conjugate).

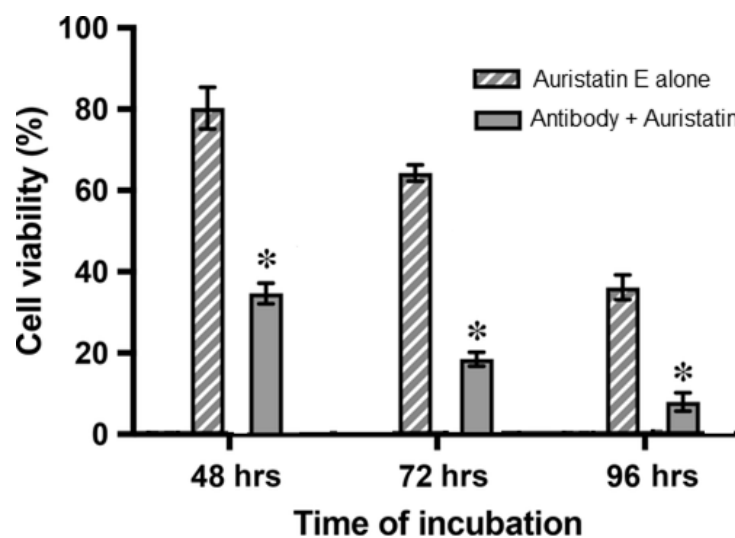
Another similar lymphoma cell line was incubated *in vitro* with auristatin E alone (unbound to antibody).

The investigation was done twice, with 9 replicates for each duration of incubation.

Cell viability (% of cells that survive) was assessed and expressed relative to untreated control in Fig. 1.2. The data for the control setup (for untreated cells) was not shown in the figure.

Data are presented as mean  $\pm$  S.D. (standard deviation).

Statistical differences between the two groups of cells are marked by asterisk ( $P < 0.01$ ).



Modified from Schonberger et. al, 2017, Journal of Cellular and Molecular Medicine

Fig. 1.2

**(iii)** Based on the information given, comment on the effectiveness of the Brentuximab vedotin (antibody and auristatin E) treatment in humans and discuss on the confidence in the presented data.

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..... [4]

- (c) Lymphocytes in the human body, have a limited lifespan. These cells are continually replaced by the division of **lymphoid** stem cells. Fig. 1.3 shows that when a stem cell divides, two daughter cells are produced. One remains as a stem cell, whilst the other divides and differentiates to form specialised cells.

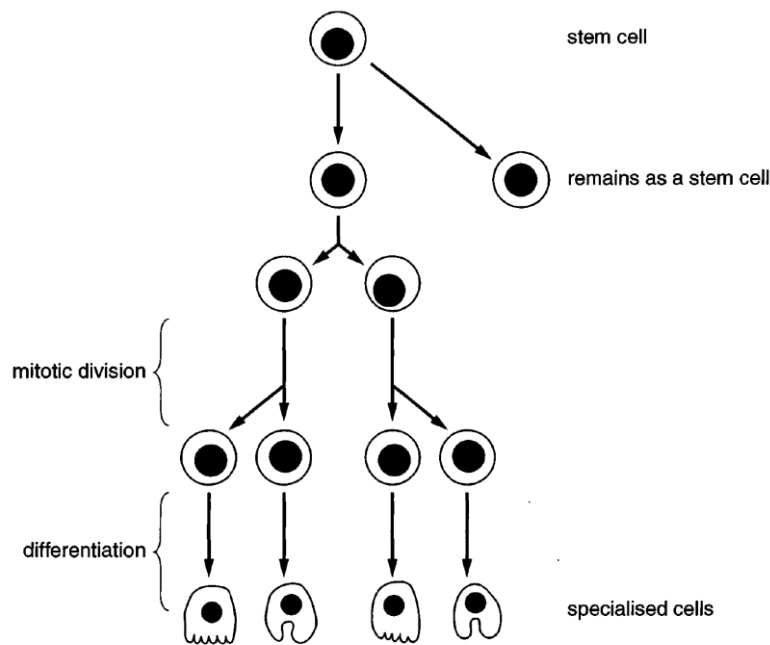


Fig. 1.3

- (i) The specialised cells are genetically identical to the stem cells from which they originated. Suggest how a cell differentiates to form a specialised cell.

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..... [2]

- (ii) Other than the location on where to find these stem cells, describe two differences between lymphoid stem cells and embryonic stem cells.

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..... [2]

**(iii)** Embryonic stem cells are extracted from human embryos 4-5 days after fertilisation, when the embryo consists of 50 to 150 cells.

Suggest one ethical concern about the use of embryonic stem cells.

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..... [1]

- (d)** Severe combined immunodeficiency (SCID) is a disease that affect the development and maturation of T and B lymphocytes.

There are two general types of SCID, both of which are caused by recessive alleles.

Fig. 1.4 shows a family pedigree including some people with a type of SCID.

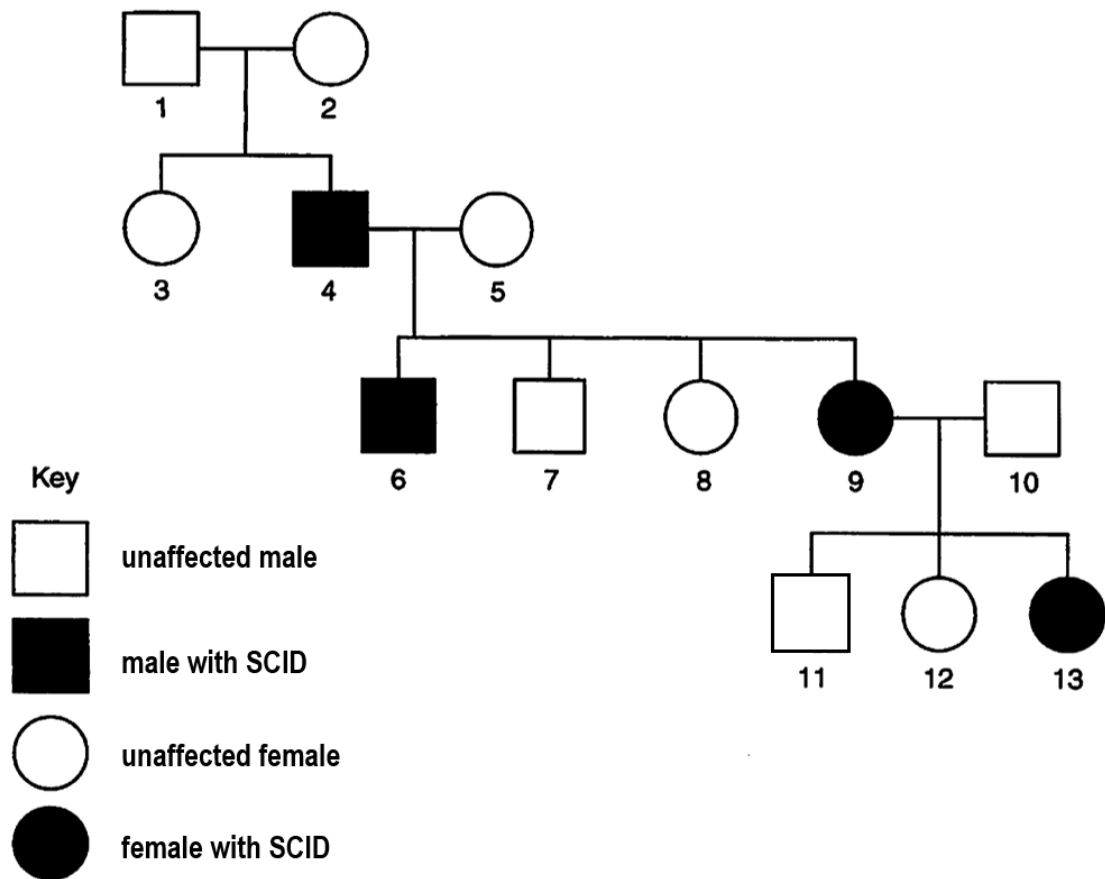


Fig. 1.4

- (i)** Explain what is meant by the term allele.

allele .....

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.....[2]

(ii) Explain, with evidence in Fig. 1.4, whether the disease is X-linked.

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.....[3]

A southern blot analysis was performed to investigate the type of alleles of this gene which causes SCID in a family branch consisting of individuals 1, 2, 3 and 4.

Fig. 1.5 shows the steps involved in this analysis.

**Steps of Southern Blot analysis**

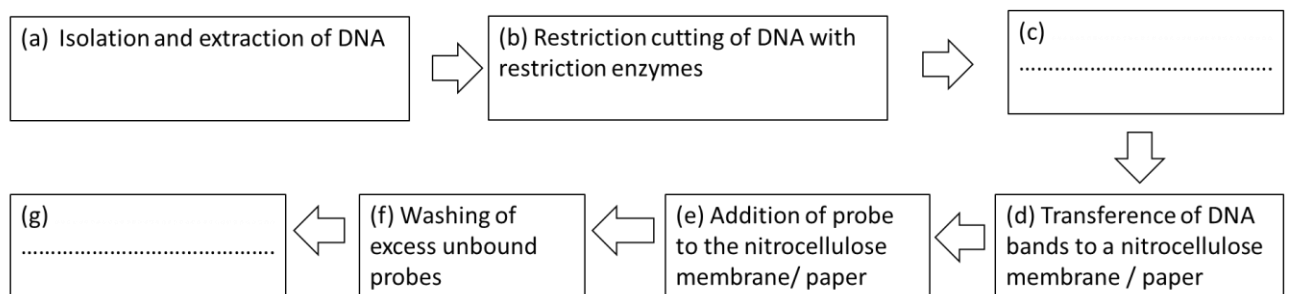


Fig. 1.5

(iii) Fill in the blanks for steps (c) and (g) as shown in Fig. 1.5.

.....[1]

Step (c): .....

Step (g): .....

Fig. 1.6 shows the restriction map of the dominant allele and recessive allele. Arrows represent restriction sites which restriction enzyme *BamHI* recognises and binds. The size of the DNA fragments after restriction cutting is indicated in kb.

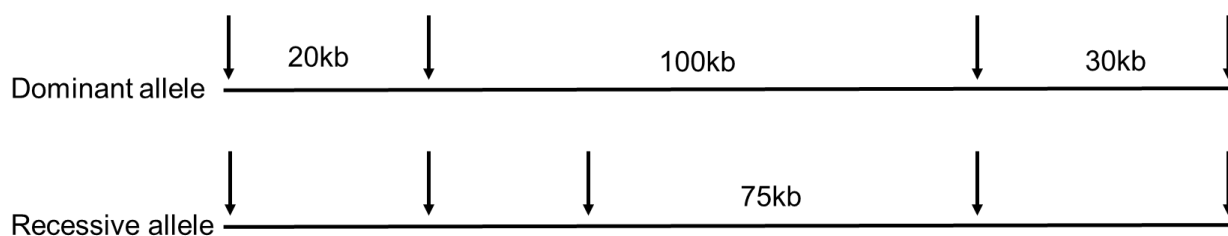


Fig. 1.6

A zoomed in view on each restriction site shows the following in Fig. 1.7.



Fig. 1.7

**(iv)** With reference to Fig. 1.6 and Fig. 1.7, describe and explain the mode of action of restriction enzyme *BamHI*.

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.....[3]

**(v)** Compare between probes (an example is the one used in step (e)) and primers used in Polymerase Chain Reaction (PCR).

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.....[2]

**(vi)** Suggest the significance of step (f) in Fig. 1.5.

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.....[1]

Fig. 1.8 shows the autoradiogram in step (g).

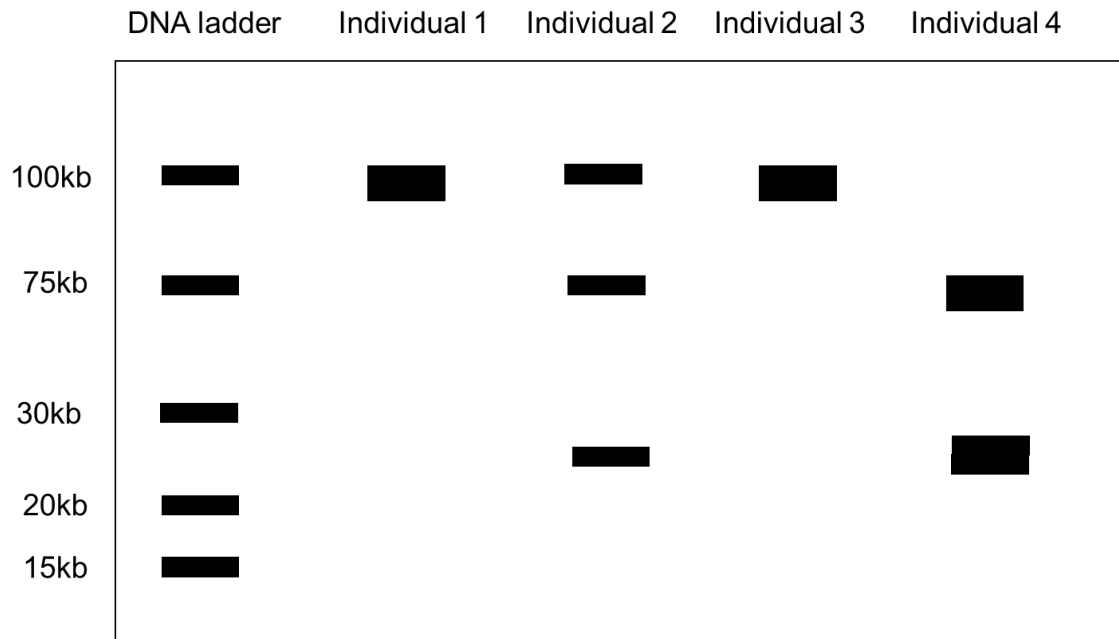
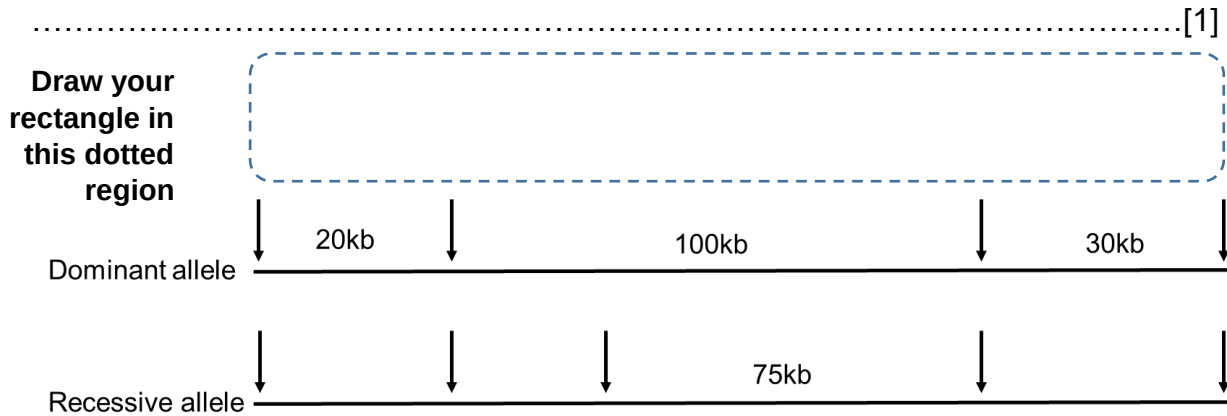


Fig. 1.8

(vii) With reference to Fig. 1.8, and using a rectangular box (  ) to represent the probe, suggest a suitable region which the probe is complementary to.

Draw your rectangular box in the **dotted region** of the diagram below.



**(viii)** With reference to Fig. 1.4 and the results of the autoradiogram in Fig. 1.8, explain in detail, how this condition might have arisen in individual 4.

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.....[3]

**(ix)** Individual 12 settled down with a male with no family history of SCID.

Calculate the probability of this couple having a son who is a carrier of the disease. Show your working clearly.

.....[2]

Probability = .....

**[Total 35 marks]**

## QUESTION 2

(a) Researchers of the International AIDS Vaccine Initiative (IAVI) and Scripps Research have tested a new vaccine approach designed to prevent HIV infections.

The vaccine consists of the engineered HIV Envelope (Env) proteins linked to a spherical protein structure. The engineered proteins are designed to engage with specific B lymphocytes of the immune system to produce a certain type of antibody that is capable of neutralizing a wide range of HIV variants.

(i) Classify the type of immunity conferred by this mode of vaccination in table 2.1 below.

.....[1]

Table 2.1

| Type of immunity | ✓ if applicable |
|------------------|-----------------|
| Passive          |                 |
| Active           |                 |
| Natural          |                 |
| Artificial       |                 |

(b) Naïve B lymphocytes express **membrane-bound** immunoglobulin M (IgM) in the form of B cell receptors, while plasma cells express the **secreted** form of IgM. The secreted IgM lack the transmembrane domain which embeds the IgM / B cell receptor in the cell surface membrane.

These structural differences is due to the process of alternative splicing of the same pre-mRNA coded by the same gene.

Both the B cell receptor and the IgM secreted by the same cell line should recognise the same antigen shape.

Describe how the plasma membrane of naïve B cells hold onto the B cell receptors.

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(c) Indicate true or false for each of the following statements.

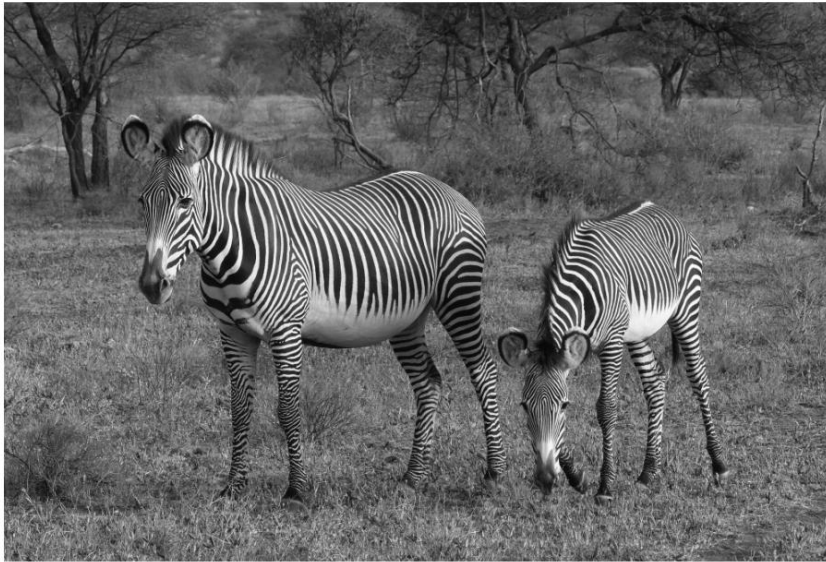
.....[2]

| No. | Statement                                                                                                                                              | True/False |
|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------|------------|
| 1   | Somatic hyper mutation occurs to allow higher affinity to the same antigen                                                                             |            |
| 2   | Gene segments (e.g. v, d, j segments) coding for variable regions of antibodies are only found in the genome of B lymphocytes.                         |            |
| 3   | <i>Mycobacterium tuberculosis</i> is a bacterium pathogen which infects the macrophages in the alveoli after transmission through air-borne particles. |            |
| 4   | Primary infection only involves the innate immune system; secondary infection involves the specific immune system.                                     |            |

**[Total 5 marks]**



**(b)** The photograph shows two zebras.



(Source: Caroline Wilcox)

In the wild, the stripes of zebras are thought to reduce attack by biting flies.

Biting flies cause stress and disease to cattle (e.g. cows), reducing food production and causing financial losses for farmers.

An investigation was carried out to see if the presence of stripes on cattle reduced the number of biting flies on their bodies.

Three groups of black cattle were compared:

- unpainted cattle
- cattle painted with black and white stripes
- cattle painted with black stripes only

Cattle repel flies by flicking their tails, stamping their feet and skin twitching.

The frequencies of these fly-repelling behaviours were recorded.

The graph (Fig 3.1) and Table 3.1 show some of the results of this investigation.

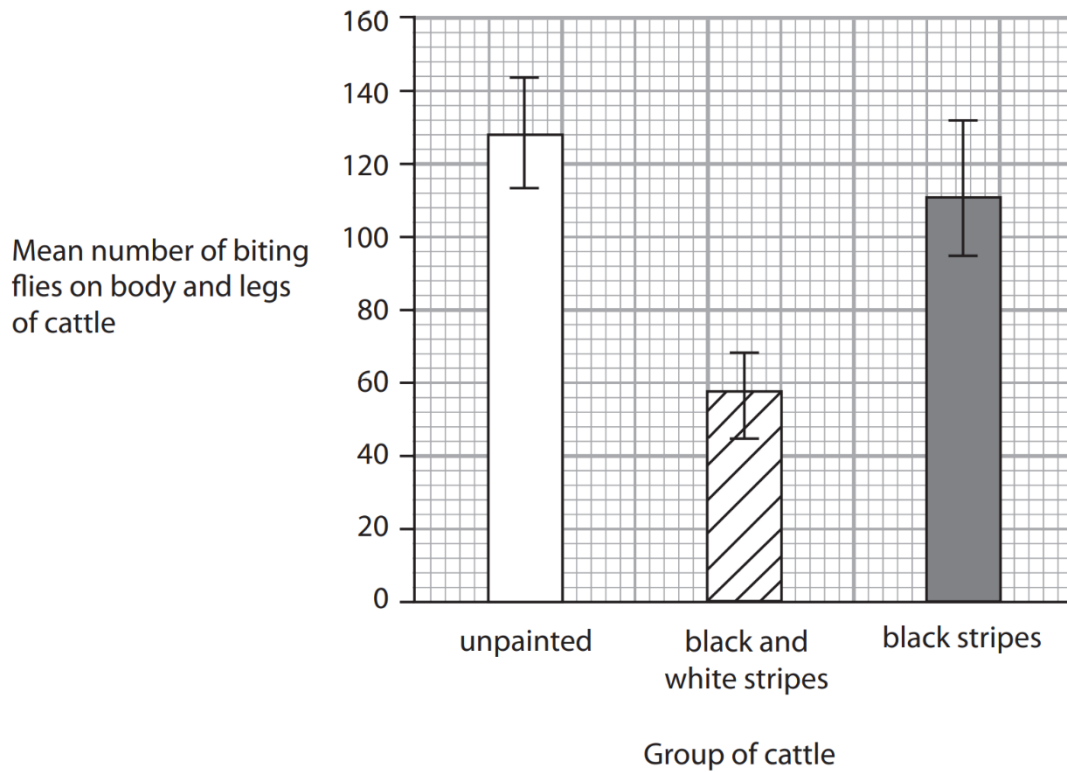


Fig. 3.1

Table 3.1

| Group of cattle         | Mean frequencies of fly-repelling behaviours / number per hour |                     |                |
|-------------------------|----------------------------------------------------------------|---------------------|----------------|
|                         | Flicking their tails                                           | Stamping their feet | Skin twitching |
| unpainted               | 54                                                             | 16                  | 5              |
| black and white stripes | 42                                                             | 10                  | 8              |
| black stripes           | 55                                                             | 16                  | 5              |

(i) Determine the effectiveness of painting stripes on cattle in reducing the number of biting flies on them.

Use the information in the graph (Fig. 3.1) and Table 3.1 to support your answer.

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.....[4]

**(ii)** Suggest a rationale for painting black stripes on the black cattle in the investigation.

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.....[1]

**[Total 10 marks]**

**Essay**

Answer **one** question only in this section.

Write your answers on the lined paper provided at the end of this question paper.

Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.

Your answers must be in continuous prose, where appropriate.

Your answers must be set out in sections (a), (b) etc., as indicated in the question.

- 4 (a)** Describe how the level of mRNA of a particular gene across different cell types in an organism is controlled. Suggest the advantage(s) of each level of control. [12]

- (b)** Discuss how complementary base pairing, both within and between biomolecules, enables prokaryotic cells to carry out processes essential to life [13]

**[Total: 25]**

- 5 (a)** Many different polymers are formed in cells. A polymer is a chain of monomers. [12]

Describe the polymerisation of different types of biomolecules in a plant.

- (b)** Discuss how the complementary shapes between biomolecules enable eukaryotic cells to carry out their diverse processes [13]

**[Total: 25]**

This image shows a full page of white paper with horizontal dotted lines, typical of primary school writing paper. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

[illegible]

This image shows a full page of white paper with horizontal dotted lines, typical of primary school writing paper. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

This image shows a full page of primary-ruled paper. It features approximately 20 horizontal dotted lines spaced evenly down the page, providing a guide for handwriting practice. The background is white, and there are no margins or other markings present.

This image shows a full page of white paper with horizontal dotted lines, typical of primary school writing paper. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

[illegible]